

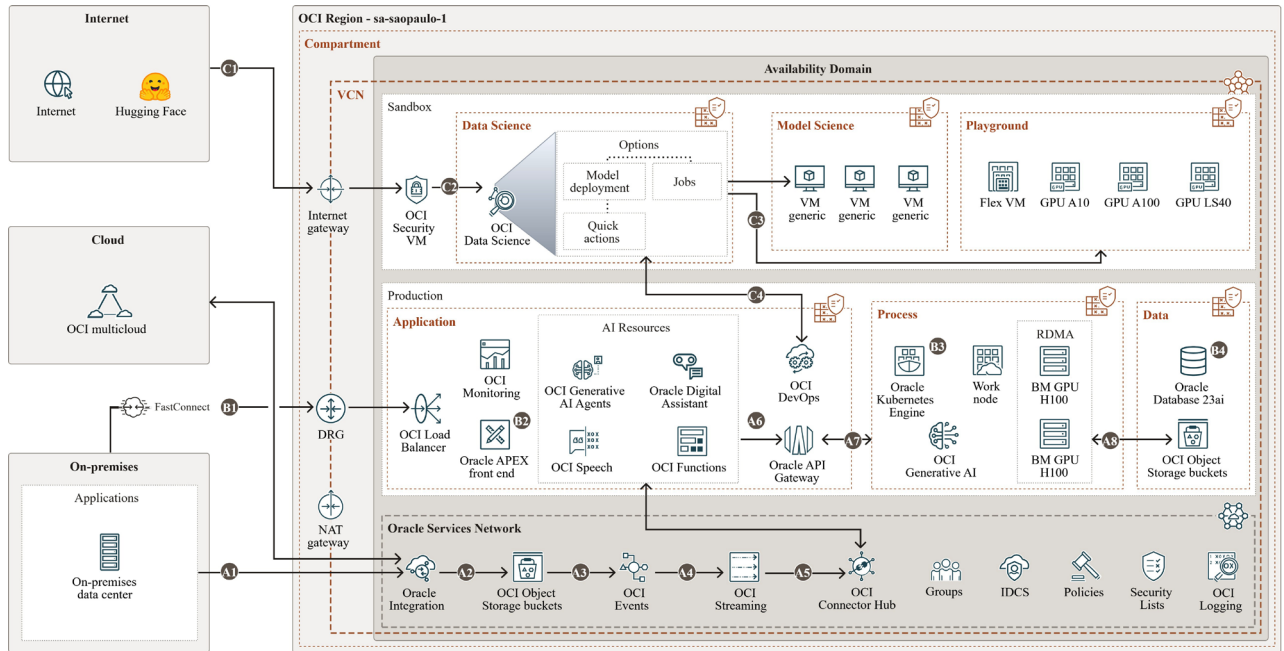


Figure 3: Oracle Identity Cloud (OCI) using its own LLM models

addition to compute services, it is also important to have efficient services for serving models. For example, vLLM and Ollama are models that make it easy to deploy LLMs by optimising the inference pipeline. They ensure that models are loaded into memory and requests are batched appropriately, and that memory is utilised efficiently within the GPU.

Observability is also an important component of an efficient AI stack. For instance, the Prometheus tool is helpful in monitoring different metrics in a system. Additionally, tools like Grafana are important for visualisation. These tools are important when dealing with large-scale AI services to avoid cost inefficiencies.

An example of an open source AI cloud stack could be:

- Kubernetes, an orchestration tool
- GPU-enabled compute nodes, which can be utilised for training and inference
- Distributed storage, which can store data and models
- Model serving frameworks, like vLLM
- Observability tools, like Prometheus and Grafana

All these tools are essential for creating an AI platform that is efficient in supporting LLM workloads. One of the most important aspects of using an AI platform is that it is customisable.

Quantum Computing as a Cloud Service (QCaaS): Architecture and integration

Quantum computing has greatly changed the way we approach problem-solving. Quantum computers have quantum bits or qubits, which can be in more than one state at a time and can perform calculations that are impossible by other means. However, the problem with quantum computers is that they are very sensitive and complicated machines and must be maintained at a temperature that is close to absolute zero. It has become impossible to put quantum computers in a data centre.

Quantum Computing as a Service, or QcaaS, is a service that provides users with the ability to access quantum computers remotely via software. Instead of developers needing to have quantum computers, they can simply send in their quantum

circuits and computations, and these will be run on a remote platform. One of the most popular quantum computing development frameworks is Qiskit. The framework is useful for developing quantum computing applications as well as for performing experiments using quantum hardware that is accessible through the cloud. For example, IBM Quantum offers quantum processors that are accessible through the cloud. This enables users around the world to perform experiments on quantum algorithms. Similarly, quantum computing services such as Amazon Braket offer unified interfaces that allow users to access multiple quantum hardware providers. With the use of these quantum computing services, users will be able to develop hybrid computing applications that leverage classical and quantum computing.

The architecture that can be expected from a typical QCaaS platform includes:

- *User interface layer*: This includes development environments and SDKs that can be used to write quantum programs.